JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 94, NO. B6, PAGES 7343-7363, JUNE 10, 1989

THE LATE PERMIAN/EARLY TRIASSIC MAGNETIC POLARITY TIME SCALE AND PLATE MOTIONS OF SOUTH CHINA

M. Steiner

Department of Geology and Geophysics, University of Wyoming, Laramie

J. Ogg

Department of Earth and Atmospheric Sciences, Purdue University, West Lafayette

Z. Zhang and S. Sun

Institute of Geological Sciences, Chinese Academy of Geological Sciences
Beijing, People's Republic of China

Abstract. Fossiliferous stratigraphic sections spanning the Permian-Triassic boundary in south China have yielded a well-defined magnetic polarity sequence for the Lower Triassic and a more poorly defined sequence in the Upper Permian. Four sections comprising more than 1100 continuous meters of stratigraphic section display a sequence of 11 major reversals of polarity within the Early Triassic (Griesbachian to Spathian stages) and a minimum of two periods of normal polarity in the Late Permian (Changsing and Wujiaping stages). Synthesis of all Permo-Triassic studies indicates that the lengthy late Paleozoic interval of reversed polarity ended after about two thirds of Permian time had passed. The record of frequent changes of polarity of the geomagnetic field during the Early Triassic is the same record that has been derived from many terrestrial redbed sequences. There is notable agreement in polarity pattern between the magnetostratigraphic-biostratigraphic record of this study in marine rocks and those of the redbed sequences, indicating that some redbeds may record geomagnetic field history with great accuracy. The similarity of the records implies that a true record of geomagnetic polarity has been obtained for the Early Triassic. The reversal frequency during the Early Triassic was one to three reversals per million years, depending on the actual duration of this time period. Three latest Permian-Early Triassic paleopoles were derived from this study, which in combination with those of earlier studies indicate that late Paleozoic to present apparent polar wander for the South China Block involved a relatively simple northward motion and that no movement of the South China Block occurred throughout the Triassic. A combined paleopole for the latest Permian-Triassic of south China is 223.6°E, 46.1°N (N=4,k=1161.2, α95=2.7).

Introduction

The Boundary Between the Paleozoic and Mesozoic

This boundary records a most dramatic extinction of life forms, representing the greatest destruction of the marine ecosystem within the geologic record. Approximately 50% of preservable marine families and between 77 and 96% of the marine invertebrate species were affected [Sepkoski, 1982]. Devastation of the marine fauna is accompanied by a significant decrease in terrestrial reptilian faunas [Olsen, 1982].

The Permo-Triassic boundary is normally a stratigraphic unconformity around the globe; some possible exceptions are sections in Greenland, on the Iran/USSR boundary, in Kashmir, and in south China [Kummel and Teichert, 1973; Tozer, 1973; Newell 1978; Nakazawa et al., 1980]. In south China, numerous sections contain the boundary, and these appear to be some of the most faunally complete in the world [e.g., Tozer, 1979]. These strata contain diagnostic Late Permian and Early Triassic conodonts, ammonites, and bivalves. Recently, two of these sections containing the boundary have been reported to contain anomalous concentrations of iridium and other siderophile metals within a boundary clay layer that is widespread over south China [Xu et al., 1984, 1985; Sun et al., 1984a,b]. The association of anomalous elemental concentrations with massive extinctions, similar to the Cretaceous/Tertiary boundary, has been inferred to indicate exterrestrial influence in this major extinction.

The apparently uninterrupted sections in south China, indicated both by the faunal completeness as well as the reported presence of an iridium anomaly, the preservation of which also suggests a lack of missing time, has led to their consideration as candidates for the global Permo-Triassic boundary stratotype. This apparent continuity, coupled with the similarities to the Cretaceous/Tertiary boundary, prompted us to obtain a record of geomagnetic polarity in these sections across this geologic event.

The Question of an Iridium Anomaly

Among the sections we sampled was one of those reported to contain an iridium anomaly (Shangsi). In addition to paleomagnetic samples, samples were also collected for a verification of the previous iridium analysis. Splits from the original material from which the iridium anomaly was observed also were kindly provided by Xu D.-Y. All samples were analyzed for iridium by F. Asaro. Although siderophile elements do appear to be present in anomalous concentrations, no amount of iridium above background levels was observed [F. Asaro, personal communication, 1986]. Similarly, analy-

Copyright 1989 by the American Geophysical Union.

Paper number 89JB00045.
0148-0227/89/89JB-00045$05.00

ses by another laboratory of samples from the other section from which an anomalous concentration of iridium had been reported (Changsing/Meishan) also found no anomaly [Clark et al., 1986]. Hence we now tend to regard the existance of anomalous concentrations of Ir in south China with skepticism.

Geology

Lithology

In Sichuan Province, the Late Permian consists entirely of limestones. In the north, lying on the Early Permian Maokou Formation are massive, dark gray, thick-bedded limestones with abundant chert, the Wujiaping Formation (Wujiaping Stage). The Wujiaping is overlain by thinner bedded, very dark gray, siliceous carbonaceous limestones and shales of the Dalong Formation (Changsing Stage). In southern Sichuan, massive gray to light gray micritic limestones of the Changsing Formation constitute the uppermost Permian strata.

A lithologic change between Late Permian and Early Triassic strata occurs everywhere in China, a global feature of all P/Tr boundary sections. This change may suggest a hiatus of unknown duration. In Sichuan the Late Permian limestones are conformably overlain by a very shallow marine to tidal flat sequence of Early Triassic age, the Feixianguan Formation. Thin bedded, evenly bedded, gray and purple limestones alternate with purplish-red and gray calcareous siltstones, micrite, and dark red shale. The formation has four members of alternating silty and limestone facies. The Feixianguan Formation is about 400 m thick in southern Sichuan [Wang et al., 1979].

The Feixianguan Formation is conformably overlain by limestones of the Jialingjiang Formation. There is little indication of a break in sedimentation, the main changes being a slight change in facies and loss of any reddish coloration. The Jialingjiang Formation also consists of four members of two alternating facies. The lithologies are gray thin-bedded limestone and gray and brown dolomite containing some collapse breccias resulting from the dissolution of evaporites. The upper (dolomite) member is almost entirely solution breccias and contortedly bedded dolomites, and hence is excessively vuggy. The total thickness in southern Sichuan is approximately 600 m [Wang et al., 1979]. The Jialingjiang Formation is disconformably overlain by the Middle Triassic (Anisian?) Leikoupo Formation.

Age

Permian. The Wujiaping Formation contains fusulinids, other foraminifera, conodonts, and brachiopods which define the Chinese Wujiapingian stage of the Late Permian [Li et al., 1986]. Global correlation within the Late Permian is frustrating at best: the Wujiaping Stage may correspond to the lower two thirds of the Dzhulfian Series, the Araksian and Chhidruan stages [Wang et al., 1987]. In North American stratigraphy, this may mean Ochoan and possibly uppermost Guadalupian Series. In Russian terminology, the Wujiaping falls somewhere within the Tartarian Stage. Where we sampled the formation, it overlies black siliceous limestones of the Maokou Formation, whose age is by definition the Maokou Stage, considered Early Permian in the Chinese system [Li et al., 1986; Wang et al., 1987].

The Dalong Formation overlying the Wujiaping Formation has numerous ammonites, conodonts, brachiopods, radiolarians, and bivalves [Li et al., 1986]. The lower member is of Wujiaping age, while the upper member represents the Changsing Stage. The Changsing Stage has been suggested to be the youngest Permian known; it may be the younger than Late Permian elsewhere or may be equivalent to the Dorashamian Stage [Kozur, 1978; Tozer, 1979], or the uppermost part of the Dzhulfian Series [Wang et al., 1987]. The relationship of the Changsingian Stage to the North American Permian is unclear; it may encompass part of the Ochoan or be entirely younger than Ochoan and probably is equivalent to the upper part or all of the Late Tartarian of the USSR.

The Changsing Formation of southern Sichuan is less well studied but apparently contains fusulinids, other foraminifera, brachiopods, and corals which suggest its age to be Changsingian Stage [Zhu et al., 1984].

Triassic. The lowest 100 m of the Feixianguan Formation contain a conodont fauna diagnostic of the Griesbachian Stage, the earliest stage of the Early Triassic. However, species representative of the late Griesbachian (particularly Isarcicella isarcicus [Sweet, 1979, also personal communications, 1986, 1987]) are found within 10 m of the P/Tr boundary in northern Sichuan and variously reported at either 0.2 or 70 m above the boundary in southern Sichuan [Wang et al., 1979; J. Yao, personal communication, 1986]. This suggests that the early Griesbachian is either condensed or missing or possibly is a nonexistent time period. The overlying 300 m of the Feixianguan Formation contain largely pelecypods and are barren of conodonts. However, they are conformably overlain by the Smithian-Spathian Jialingjiang Formation; their considerable thickness and this stratigraphic position suggest that these beds represent the Dienerian and possibly early Smithian stages, the second and third stages of the Early Triassic.

The Jialingjiang Formation contains conodonts throughout the four members, which are known to range through the Smithian and Spathian (third and fourth stages of the Early Triassic). However, most of the species are long-ranging, and therefore, while the fauna is sufficient to indicate the presence of these two stages in the Jialingjiang Formation, it is inadequate to define a boundary between them or to determine the extent of the Spathian present.

Sampling

Four stratigraphic sections were sampled. Two of the localities were exposures of appreciable amounts of both Late Permian and Early Triassic strata and were supposedly continuous across the boundary. The third locality presented nearly the entire Early Triassic, and the fourth, nearby to the third, was sampled to obtain a fold test within the Triassic strata. All sections are in areas which were gently folded at the end of the Late Cretaceous or earliest Tertiary [Yang et al., 1986; Xiao X., personal communication, 1987].

The most complete and well-studied section across the boundary is that near the village of Shangsi (106.4°E, 32.4°N), about 70 km southwest of the city of Guangyuan in northern Sichuan (Figure 1). The location of the Permo-Triassic

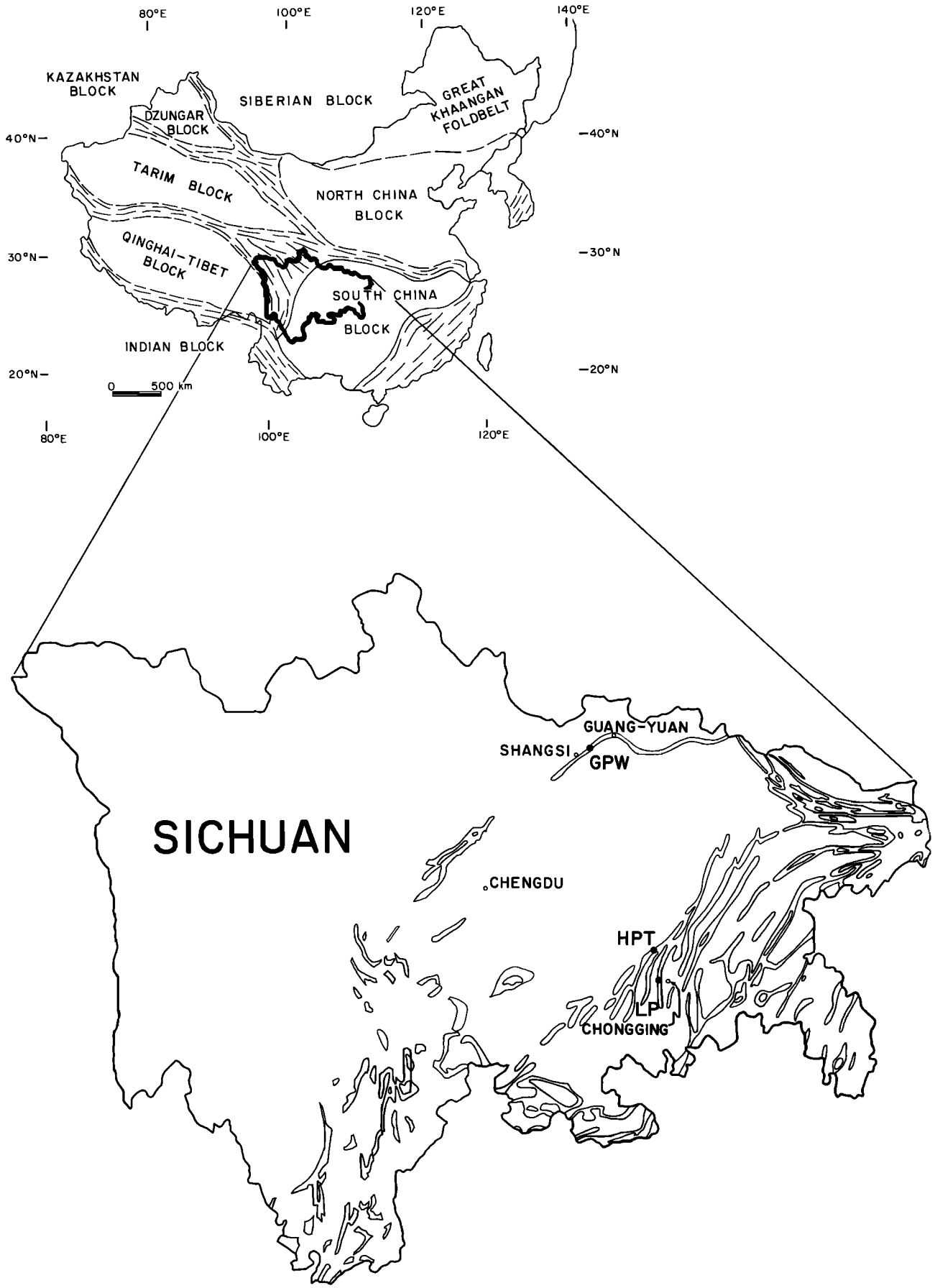


Fig. 1. (Top) Tectonic provinces of China; (bottom) location of the sampling sites, and outline of Late Cretaceous/Early Tertiary folds exposing Mesozoic strata.

boundary is marked by faunal and lithologic changes and by a clay layer from which the anomalous iridium and siderophile elements have been reported. The strata are exposed in a roadcut and dip 50° to 60° to the SE. The well-exposed Early Permian Makou Formation at the base of the exposure was not sampled because of its dark gray to black coloration. Paleomagnetic sampling began at the base of the Wujiaping Formation and spanned the entire Wujiaping and Dalong formations (the entire Late Permian of Chinese terminology) at about 1-m intervals. Continuing into the Triassic with the same sampling density, the sampling encompassed three fourths of the lower member of the Feixianguan Formation. A total of 236 samples were taken over 198 m at this "GPW" section.

The second boundary section was located in southern Sichuan west of Chongqing (Figure 1) near the village of Liangfenya (105.6°E, 29.6°N). The exposure is a small limestone quarry. Beds dip about 55° to the west (N75°W). Here the Permian limestones had been altered from gray to a grayish yellow. This section, "LP," sampled 91 m of the upper Changsing Formation and the overlying 21-m exposure of the lowermost Feixianguan Formation. Contrary to our expectations, we found that the system boundary here is tectonic. Boudinage structures are present in the clay layer between the Permian and Triassic formations, indicating slip along this horizon during folding. Forty-eight samples were taken from the 112 m of Late Permian and Early Triassic exposed.

Ninety kilometers to the north of Chongqing (106.3°E, 29.9°N), near the town of Hechuan, a roadcut exposes the entire Feixianguan and Jialingjiang formations. Two sections were sampled on either limb of an anticline cut by the road. Sampling began just meters below the P/Tr boundary. The large section, "HPT," encompasses the the highest bed of massive Upper Permian limestone, 1-m thick, and continues through both Lower Triassic formations, encompassing most of the Early Triassic. The strata dip generally about 30°-40° to the SE (N130°E). Two hundred and seventeen evenly spaced stratigraphic levels were sampled throughout 960 m; the highest 40 m of the section were excluded because of extensive dissolution and deformation.

About 0.7 km northwest of the HPT section, the uppermost 4 m of the Late Permian, and 154 m (with a 100-m gap) of the Early Triassic Feixianguan Formation were sampled, the "HT" section. The beds dip 40°-75° to the NW. Forty-two samples were collected. A fault cuts this exposure just above the P/Tr boundary. By matching the lithostratigraphy with the HPT section, we were able to determine the exact amount of section missing as a result of faulting. The HT and HPT sections together provide the opportunity for a fold test of the remanence age, and HT further provides a duplicate section for a portion of the Triassic polarity sequence.

Treatment

Samples were measured and demagnetized within a steel magnetically shielded room. A computer-interfaced SCT cryogenic magnetometer, with a practical noise limit of 5×5^{-6} A/m (5×10^{-9} emu/cm^3), was used for all measurements. Multiple readings per measurement were taken, according to the strength of the samples. Samples were demagnetized both in alternating fields AF (using a single axis Schonsted GSD-1 AF demagnetizer) and thermally. Thermal demagnetization employed a 25-sample horizontal oven, having an ambient field in the cooling chamber of $\leq$5 nT.

Duplicate specimens of representative samples from each lithology at all localities were treated by both thermal and AF demagnetization. These pilot samples were stepwise AF demagnetized in fields of 5 to 40 mT. AF treatment produced substantial intensity loss but no significant separation of components by 40 mT; therefore demagnetization was not continued in higher fields. Duplicate specimens were then heated in 30°-50°C steps, beginning at 150°C and continuing up to 650°C. The maximum temperature depended on the lithology of the samples. Samples from Permian limestones were heated to only 400° or 440°C, as their moments approached the noise limit of the magnetometer at those temperatures. Samples from most Triassic lithologies (except recrystallized dolomites) were heated up to 630° or 650°C.

Results

The natural remanent magnetization (NRM) directions of all samples are shown in Figure 2. Considerable differences between the remanent magnetization characteristics of the Late Permian and the Early Triassic strata necessitate the discussion of each separately.

Late Permian (GPW, LP)

Late Permian limestones had NRM intensities between 10^{-4} and 10^{-3} A/m. NRM directions reflected significant secondary magnetization (Figure 2), most often associated with the present-day geomagnetic field direction but also to a lessor extent with that of the Late Cretaceous. Present-day and Cretaceous magnetizations also are those observed as removed vectors during heating to 350°C. The inference of a Cretaceous age stems from the coincidence of NRM and removed vector directions with the mean Late Cretaceous directions obtained in south China [Otofuji et al., 1983; Kent et al., 1986]. The Cretaceous secondary magnetization is particularly significant in the LP section and to a minor extent in the GPW section. Current data do not permit a distinction between Jurassic and Cretaceous poles in south China [e.g., Courtillot and Besse, 1986]; thus the overprint could be of either age.

The response of these limestones to AF demagnetization to 40 mT was generally a rapid, usually order of magnitude, decay of intensity. However, separation of secondary and characteristic components was minimal, leaving recognized secondary magnetizations as the dominant magnetization. After thermal demagnetization to up to 440°C in six to seven increments, directions in slightly over one half of the samples moved along great circle paths toward normal and reversed Permian directions. Complete separation of the remanence components usually was not achieved by 440°, after which samples either were near the noise limit of the magnetometer or had begun to exhibit significant increases in intensity, probably reflecting the production of magnetite from ferrous carbonates. The other half of the Permian sample population retained secondary magnetizations completely throughout demagnetization or to an extent

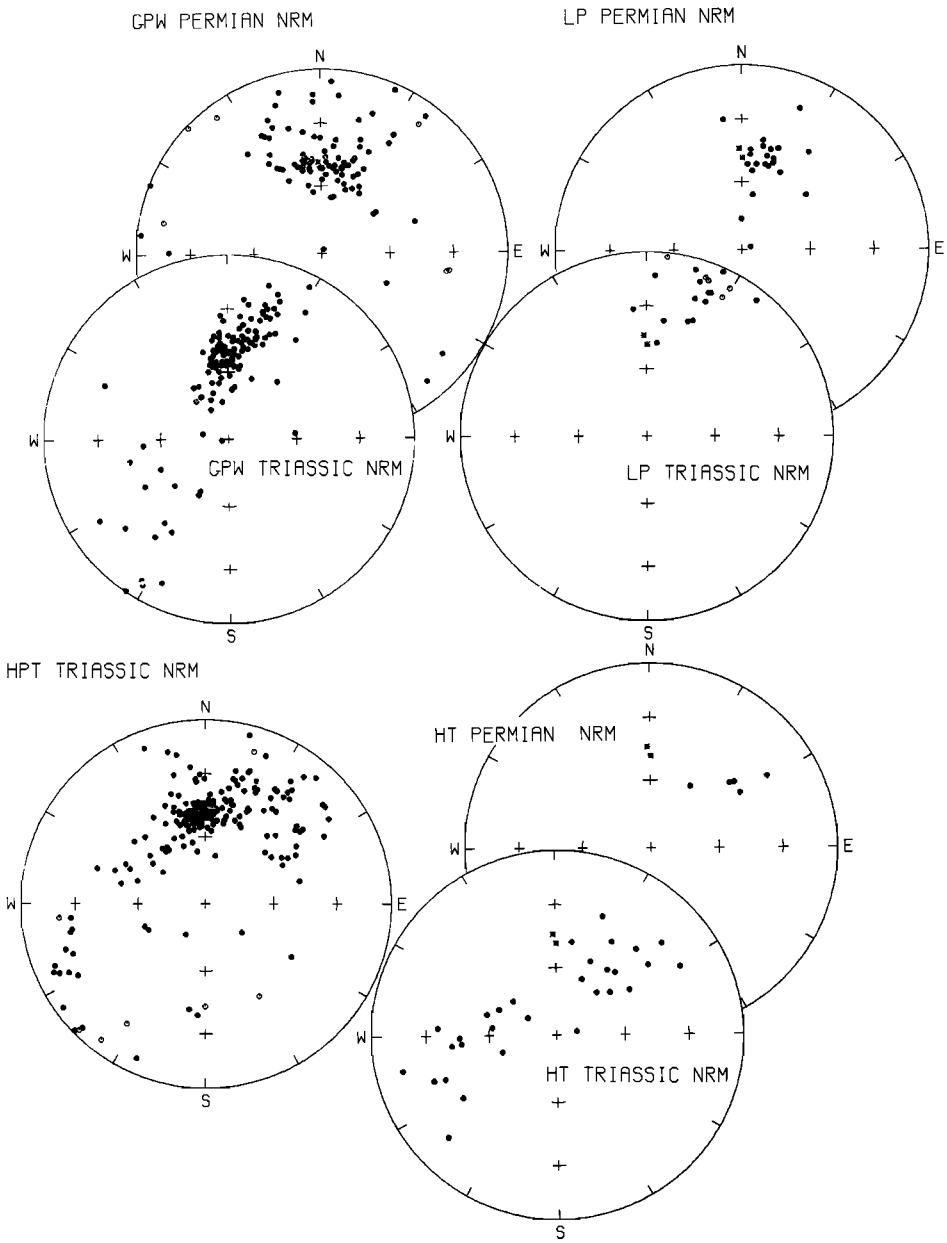


Fig. 2. NRM directions (geographic coordinates) from each locality, separated into Permian and Triassic groups. Asterisks represent the present and axial field directions. Equal-area projection; open (solid) circles represent points on the upper (lower) hemisphere.

that no characteristic polarity could be interpreted.

Examples of the demagnetization responses of the half of the Permian samples which did respond to demagnetization sufficiently to allow polarity interpretations are shown in Figure 3. Only a very few samples exhibited univectorial decay of magnetization to the origin of orthogonal axes plots as shown in Figure 3a. Behaviors illustrated by Figures 3b-3d were typical of others. Figures 3a and 3b illustrate samples with normal polarity magnetizations, while Figures 3c and 3d display inferred reversed polarity magnetizations. The intensity increases above 370°C in Figures 3b and 3c were fairly common.

Early Triassic (HPT, HT, GPW, LP)

The Feixianguan and the Jialingjiang formations contain a variety of lithologies which exhibit different demagnetization responses. AF demagnetization to 40 mT usually reduced the intensity of these samples to about 20% and resulted in univectorial decay toward the origin in some samples. As with the Permian samples, however, AF demagnetization of obviously overprinted samples yielded little significant separation of components; however, thermal demagnetization of duplicate specimens yielded clear separation. This response suggests a characteristic remanence held in magnetite with secondary overprints in a higher coercivity

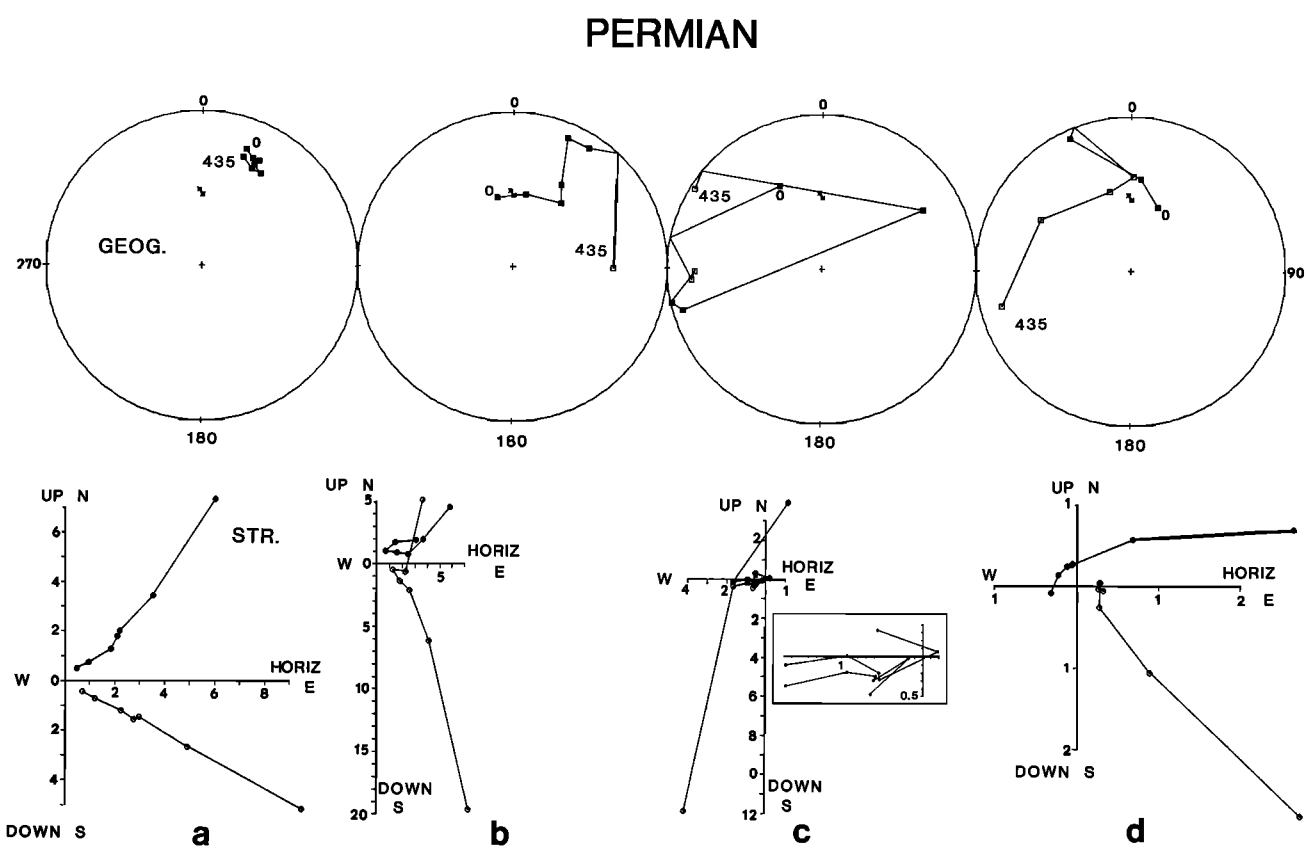


Fig. 3. Representative thermal demagnetization responses of normal [(a) and (b)] and reversed [(c) and (d)] Permian samples. Both stereographic and vector plots are shown for each sample. Stereograms show the data in geographic coordinates along with the location of the present and axial field directions; vector plots display the same data after correction for geologic structure. Temperature steps are 250°, 300°, 330°, 370°, 410°, and 435°C. Symbols in stereograms are the same as in Figure 2. Vector plots display horizontal and vertical vectors (solid and open circles) corresponding to the declination and true vertical inclination components of the magnetization vector. Scale divisions on all plots are 1×10^{-5} A/m (10^{-8}emu/cm^3).

mineral, probably goethite or maghemite which inverts to hematite on heating.

Pilot samples were heated in 13 steps from 150° to 650°C. The remainder were heated in 8-12 steps from 250° to either, 525° for those of limestone or dolomite composition, or 600°C for those of, or intimately associated with, the reddish silty facies. Demagnetization behaviors of the limestone and the silty facies are similar up to temperatures between 400° and 525°C. Typical examples are shown in Figure 4; because of the similarity of responses below 525°C, Figure 4 shows samples from facies which survive heating to 600°C.

The demagnetization responses of samples of the silty (frequently reddish colored) facies were very clear. NRM intensities ranged from 1×10^{-1} to 1×10^{-2} A/m, and demagnetization behavior was generally linear from temperatures of 450° to above 600°C (Figure 4). The only secondary magnetization observed in these samples was that in the present/axial field direction. Nearly all of the samples exhibited linear decay, usually between temperatures of 450° and 600°C. While statistical linearity in many sample directions persisted above 600°C, errors generally increased.

The magnetizations of samples of the purely calcareous compositions (limestones and dolomites) had NRM intensities of 1×10^{-3} to 1×10^{-4} A/m. These usually exhibited linear demagnetization behavior to 525°C, after which directional stability became erratic and intensities either approached the noise limit of the magnetometer or exhibited significant gains. The dolomitized samples particularly exhibit very large intensity increases and erratic directional behavior upon heating to temperatures above 450° or 500°C, probably reflecting the breakdown of ferroan dolomite and subsequent production of magnetite. Univectorial decay was exhibited between temperatures of 370° and 525°C. Throughout this population, isolated samples showed only a present field magnetization and exhibited large intensity losses on heating to 250°C. Samples from the largely dolomitized upper part of the Jialingjiang Formation, particularly above 730 m, frequently were extensively overprinted with the present field direction and exhibited only overlapping blocking temperature spectra upon demagnetization.

Nearly all of the samples from the Feixianguan and from the lower half of the Jialingjiang formations displayed well-defined normal and reversed

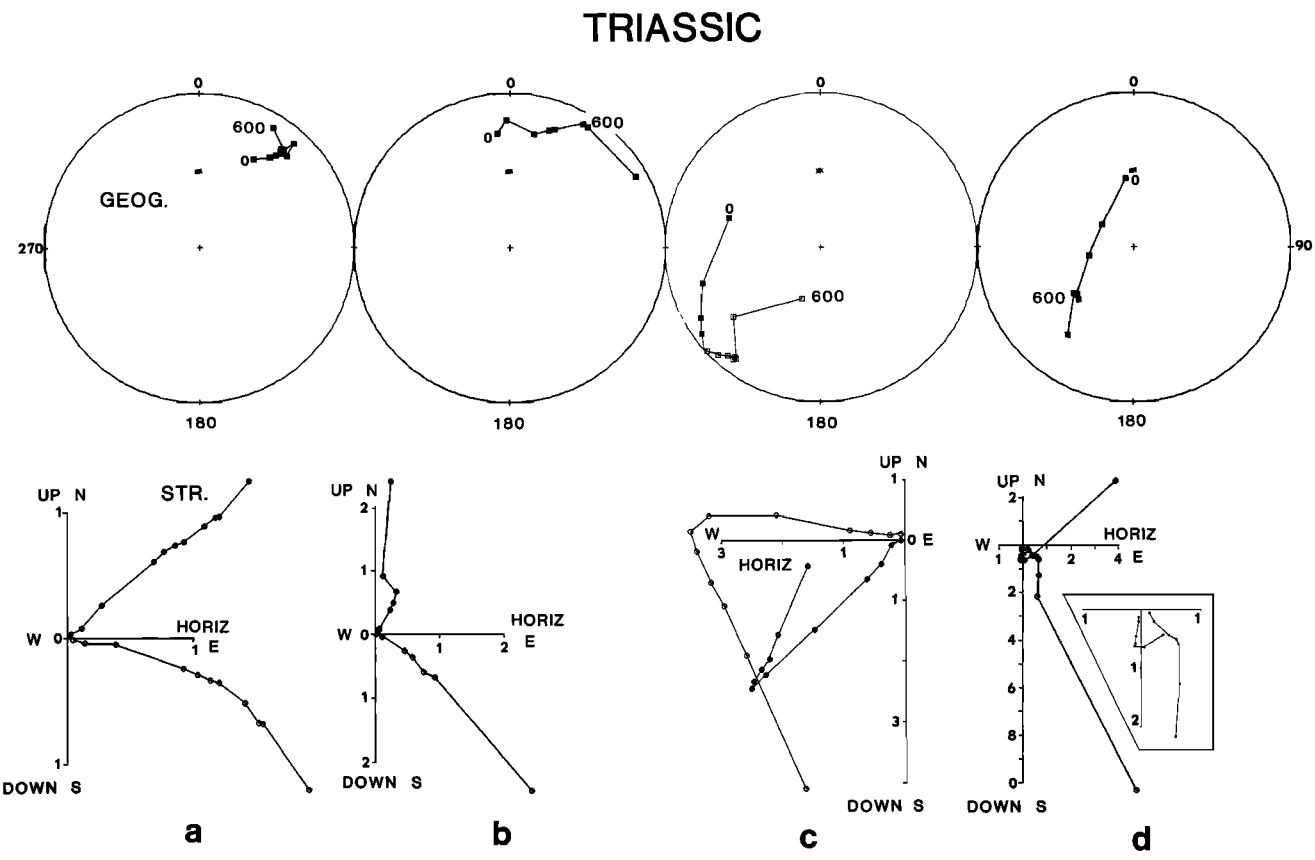


Fig. 4. Representative thermal demagnetization responses of Triassic normal [(a) and (b)] and reversed [(c) and (d)] samples. Each sample magnetization is plotted in geographic coordinates on the stereoplots and after correction for structure in the orthogonal axes plots. Plotting conventions are as in Figure 3. Temperature steps are (a) 250°, 290°, 330°, 370°, 410°, 450°, 550°, 565°, 600°; (b) 250°, 350°, 450°, 500°, 550°, 600°, 630°; (c) 150°, 180°, 210°, 250°, 300°, 350°, 400°, 450°, 500°, 550°, 600°; (d) 250°, 350°, 450°, 500°, 550°, 600°, 630°. Scale divisions are (a) $1x10^{-3}$ A/m; (b) $1x10^{-1}$ A/m; (c) $1x10^{-4}$ A/m; (d) $1x10^{-3}$ A/m.

polarity magnetizations after thermal demagnetization. Of the entire collection of Feixianguan and Jialingjiang samples, 76% displayed clearly defined polarity. Another 14% had significantly overlapped blocking temperature spectra but still afforded tentative indications of polarity which coincided with the polarity of samples above and below.

A majority of the samples from both facies show pronounced intensity decreases above 500°C. Above this temperature either decay toward the origin continued to be linear but with larger error limits or magnetization directions became altogether erratic. This behavior and the significant reduction of intensities by AF demagnetization suggest that the principal characteristic remanent magnetization carrier is magnetite in both facies.

Analysis

Directions of characteristic and secondary magnetizations and their variances were computed for all samples using a three-dimensional least squares line fitting technique [Kirschvink, 1980]. The characteristic directions are shown for each site in Figure 5. For the purposes of polarity interpretation and for inclusion in pole position calculations, samples were categorized according to their demagnetization response. Four categories evolved, of which two were used for pole calculation and three for polarity interpretation; the fourth was deemed to have no useful paleomagnetic information representative of Late Permian/Early Triassic time.

The first category consists of samples whose magnetizations are univectorial to the origin. The second category constitutes samples which were linear, but trending to points not coinciding with the origin. Third are samples which exhibit demagnetization behavior with trends similar to catagory one samples but whose overlapping blocking temperature spectra precluded separation of the magnetization components. Derivation of characteristic P/Tr magnetization directions was not possible for these samples, but they nevertheless indicate the polarity of the paleofield in which they were initially magnetized.

The fourth category consists of samples which did not exhibit Permo-Triassic directions. The direction of these samples is either that of the present-day field, the Jura-Cretaceous direction, or a direction intermediate between the Permo/Triassic field direction and one of those directions but very close to the secondary direction.

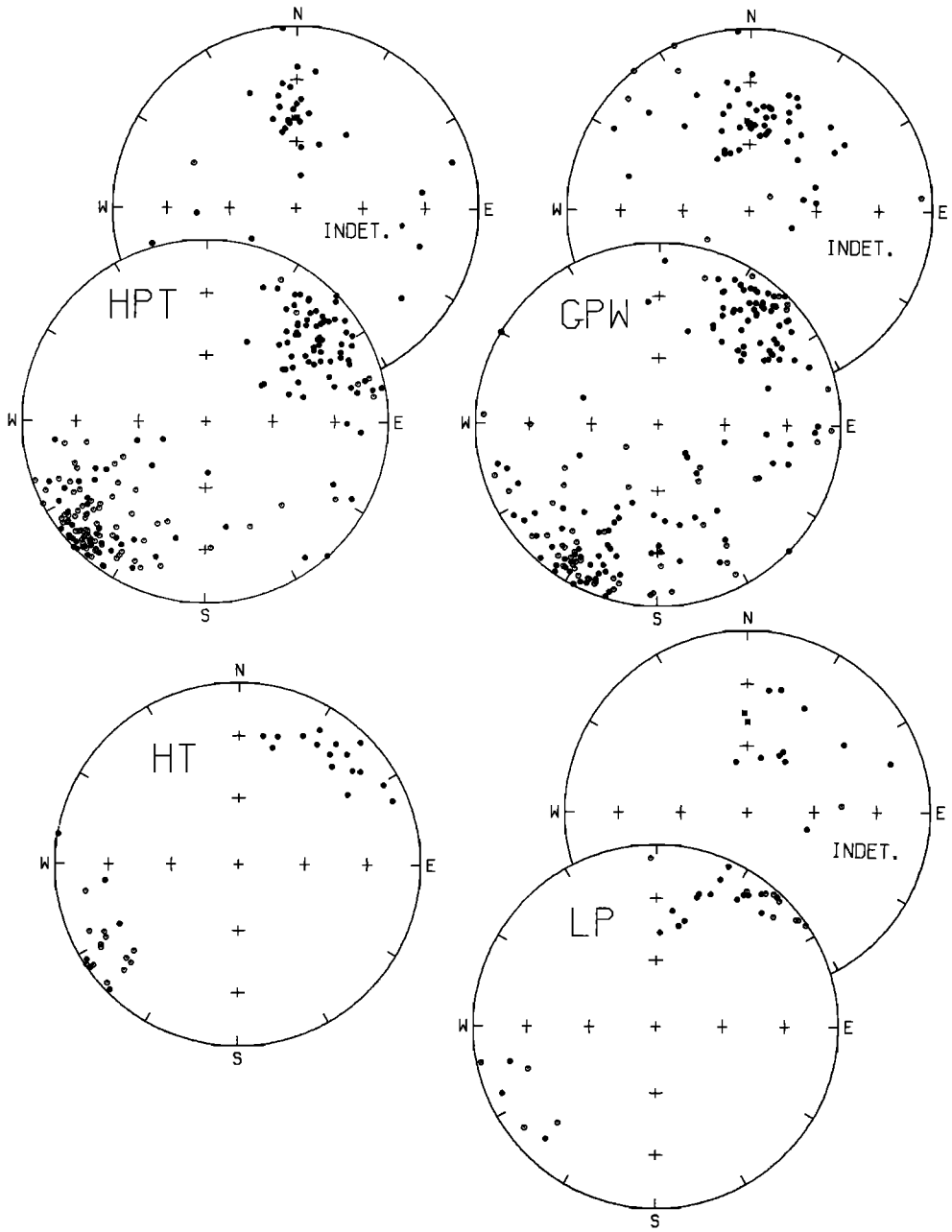


Fig. 5. Characteristic directions for all samples from each locality. Each group has been separated into plots of characteristic Permo-Triassic directions and indeterminate directions ("INDET."). Permo-Triassic directions are shown corrected for geologic structure, while indeterminate samples are shown in geographic coordinates, along with the location of the present and axial field directions (asterisks). All symbols as in Figure 1.

This fourth type is termed indeterminate. Directions of indeterminate samples are shown separately in Figure 5 and are excluded from further analysis.

Percentages of the magnetization catagories vary dramatically between Permian and Triassic strata, with Permian samples exhibiting a markedly higher percentage of the third and fourth catagories. Of a total 125 Permian samples, 42% are indeterminate, and 42% are those with significantly overlapped blocking temperature spectra, useful only for (usually) tentative polarity interpretation. By contrast, of 405 Triassic samples, only 10% are indeterminate and 14% with significantly overlapping stabilities.

Catagories 1, 2, and 3 characteristic directions are shown in stratigraphic sequence for each locality in Figures 6-9, along with polarity interpretations and lithostratigraphy. The tentative nature of the polarity interpretation derived from catagory three samples is indicated by half width bars in the polarity interpretation columns.

Two populations were used to calculate pole

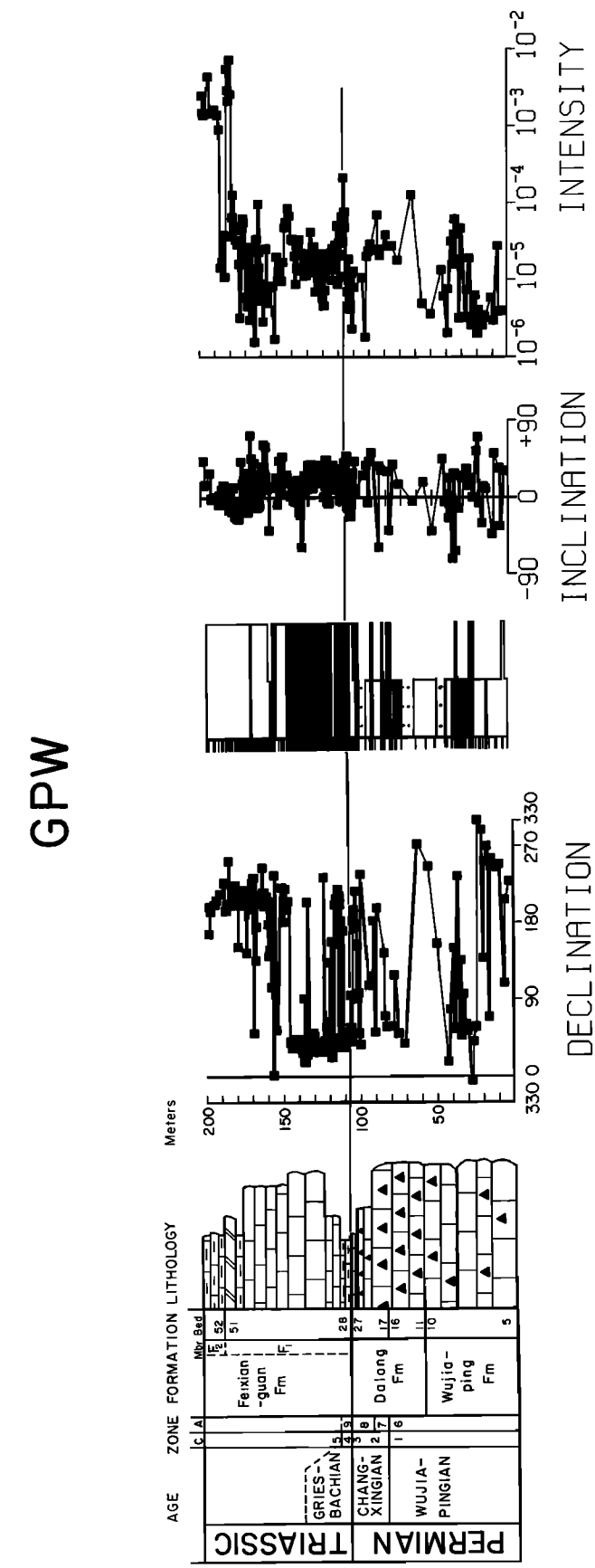


Fig. 6. Stratigraphic plot of normal and reversed directions and intensities (in A/m) for the Shangsi (Guangyuan) locality. The black/white column shows the polarity interpretation of each sample; the degree of certainty is indicated by the width of the column, half width indicating less clear demagnetization behavior, hence polarity interpretation. Single samples of polarity opposite to that their neighbors are shown in this plot as either as whole or as half bars, depending on their demagnetization responses; however, in the composite plots of Figure 12, all such single samples are depicted as half bars. Biostratigraphic zones for conodonts (C) and ammonites (A) are shown; numbers indicate the zones of 1, N. orientalis; 2, N. changxingensis; 3, N. subcarinata; 4, I. isarcica/A. parvus; 5, N. carinata; 6, Konglingites; 7, Tapashanites; 8, Rotodiscoceras/ Pleuronodoceras; and 9, Ophiceras/Hypophiceras.

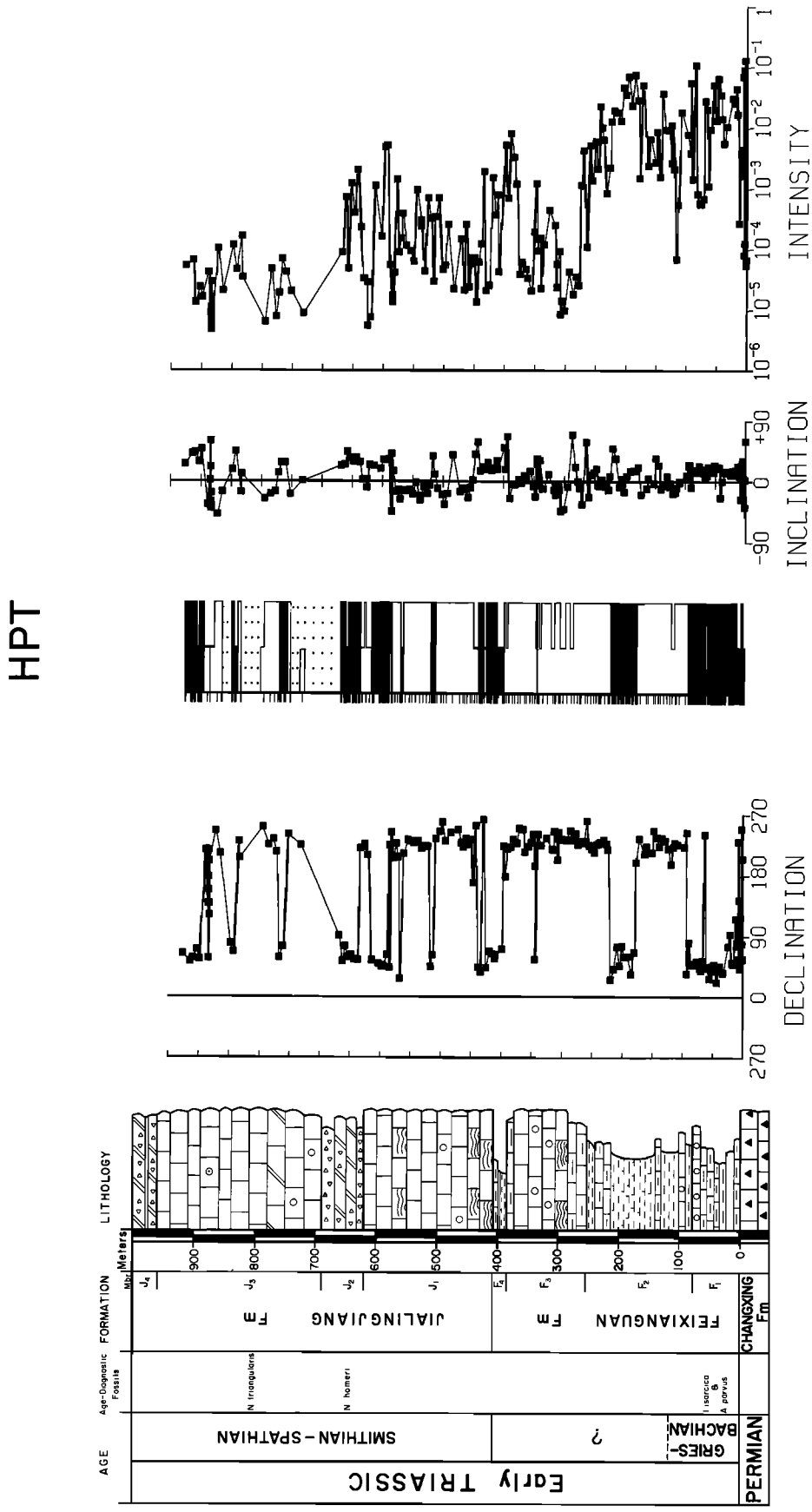


Fig. 7. Stratigraphic plot of normal and reversed directions and intensities for the southern Hechuan locality. Explanation as in Figure 6.

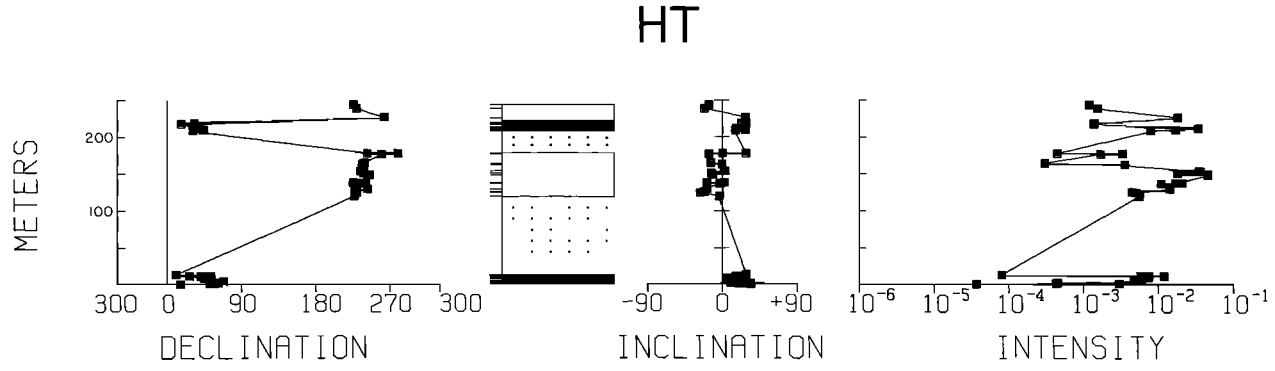


Fig. 8. Stratigraphic plot of normal and reversed directions and intensities for the northern Hechuan locality. P/Tr boundary at 4 m. Explanation as in Figure 6.

positions for each locality: one included only samples exhibiting remanence decay univectorial to the origin (category 1 samples), and a second consisted of those samples and ones with remanence decay linear to a point close, but not exactly corresponding within statistical error, to the origin (category 1 and part of category 2). The second calculation was performed to include a greater percentage of the sample population in the pole calculation in order to assess the possibility of bias originating from the smallness of the univectorial-to-the-origin sample population. No difference in directions was obtained, solely an increase in dispersion. The paleopole calculations from the first calculation are listed in Table 1.

Comparison of Paleomagnetic Poles

The apparent polar wander (APW) curve for south China was depicted by Lin et al. [1985] to move in a twisting northwesterly fashion from Carboniferous through Middle Triassic time. Subsequent studies have added new data: Zhao and Coe (1987) revised the calculation of the Emeishan Basalt pole, supposedly of Late Permian age, and new Triassic paleopoles have been provided by Opdyke et al. [1987] and by this study. The pole positions from the four localities of this study and published Carboniferous through Jurassic-Cretaceous paleopoles for the South China Block are illustrated in Figure 10.

Pole positions derived from the GPW and LP localities are quite similar. At both localities, the few Late Permian samples whose demagnetization was univectorial to the origin have the same direction as the Early Triassic samples. This suggests that the location of the pole was the same or similar in the Late Permian as in the Early Triassic.

Nevertheless, the bulk of samples contributing to the paleopoles of each locality are from Early Triassic strata. The GPW and LP poles lie close to the Early Triassic pole position obtained by Opdyke et al. [1987] (1 in Figure 10). A single Late Triassic pole [Opdyke et al., 1987] (2 in Figure 10) is also quite similar, although it was obtained from a relatively small number (23) of samples.

The paleopoles from HPT and HT are statistically identical after correction for their opposite structural attitudes (Table 2). Since these sample populations were collected from opposing limbs of an anticline, this is a positive fold test, demonstrating that the magnetization is pre-Late Cretaceous to earliest Tertiary.

The statistical coincidence of the four poles, LP, GPW, 1 and 2 strongly suggests that these determinations define the position of the Late Permian-Triassic pole for the South China Block. Although of the same age, particularly as LP, GPW, and 1, the HPT and HT poles are significantly displaced from this group (Figure 10). Their discordance in light of the coincidence of the other poles suggests that the Hechuan region is rotated relative to other parts of Sichuan and south China.

This implication of rotation of this locality was unexpected and is somewhat unfortunate. Hechuan samples responded well to demagnetization and provided a detailed polarity sequence as well as numerous univectorial samples, yielding a well-defined pole position. The normal and reversed samples contributing to this pole are very clearly

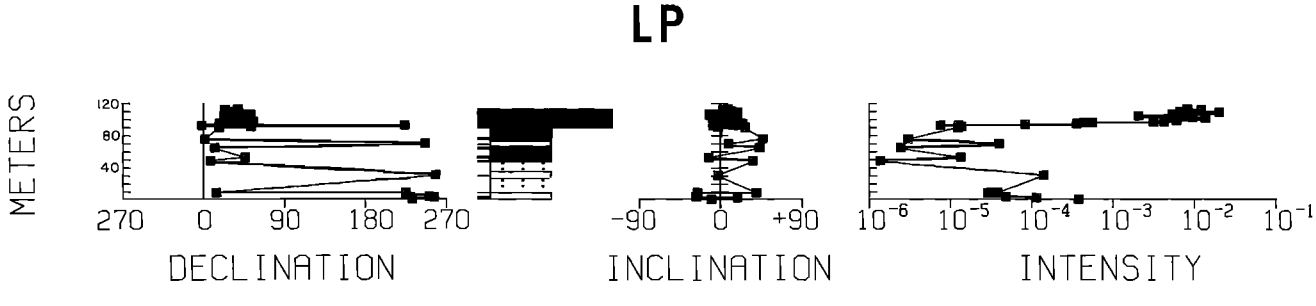


Fig. 9. Stratigraphic plot of normal and reversed directions and intensities for the Liangfenya (Chongquin) locality. P/Tr boundary at 91.6 m. Explanation as in Figure 6.

TABLE 1. Site Mean Poles

Site	Declination,deg	Inclination,deg	°E Longitude	°N Latitude	N	k	dp	dm
GPW	36.4	12.2	226.3	47.2	38	32.7	2.1	4.2
LP	7.4	6.2	225.2	45.8	17	27.5	3.5	7.0
HPT	50.4	17.1	209.8	38.5	62	33.5	1.7	3.3
HT	49.4	14.9	211.8	38.7	26	22.4	3.2	6.3

N, number of samples; k, precision parameter; dp, dm, semiaxes of ellipses of confidence.

antipodal (Table 2). The lack of appreciable secondary magnetization in most samples is attested to by a comparison between the pole computed from all samples for which normal and reverse polarity could be inferred, even including samples of extremely tentative interpretations, and the pole from only the samples whose demagnetization behavior was univectorial to the origin (Table 2).

This seeming dichotomy has prompted examination of these site locations with respect to the regional geology. The geologic map reveals that these sites are situated close to a major 30° bend

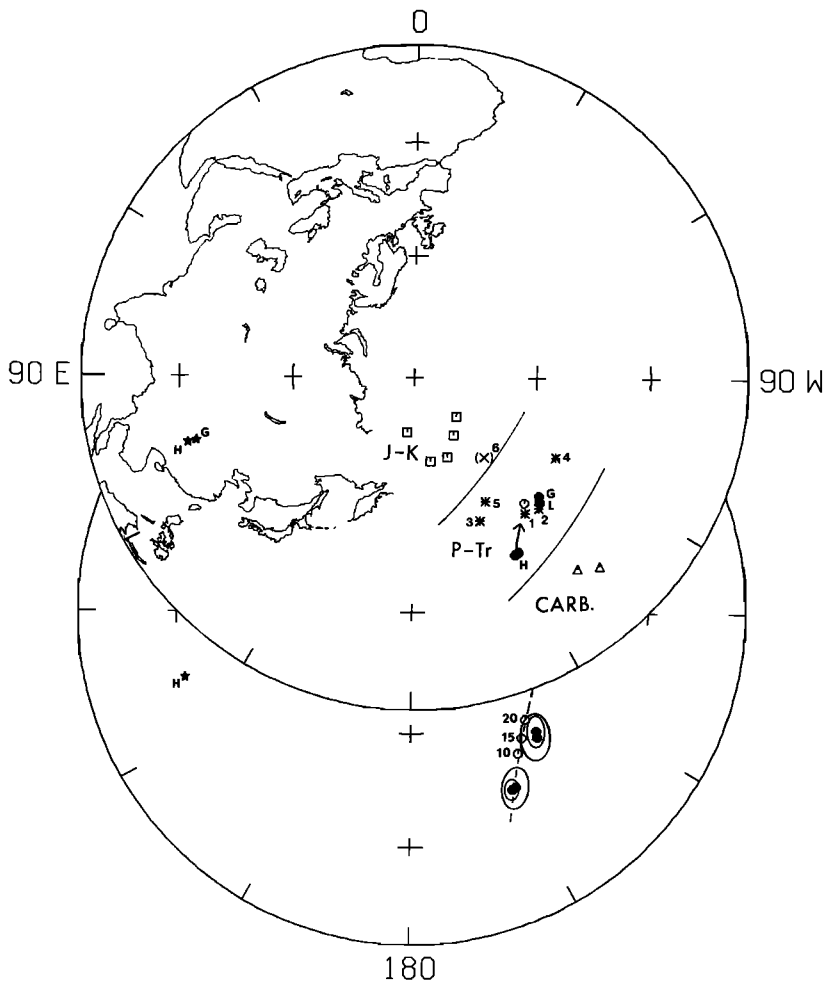


Fig. 10. (Top) Carboniferous through Jurassic-Cretaceous pole positions for the South China Block. Carboniferous poles (triangles) are those of Lin et al. [1985]. Solid dots are the poles from this study. Asterisks are published Permian-Triassic pole positions; identifying numbers are explained in the text. Jurassic and Cretaceous poles (squares) are from Lin et al. [1985] and Kent et al. [1986]. Curved lines separate poles of different age groups, depicting the relatively uniform northward motion of the pole since late Paleozoic time. The locations of the Hechuan and Guangyuan localities are indicated by stars. (Bottom) Pole positions from this study and their 95% ovals of confidence. Open circles show rotation of the HPT/HT poles by varying degrees (10°, 15°, and 20°).

TABLE 2. Fold, Reversal, and Secondary Contamination Tests at Hechuan

	Declination,deg	Inclination,deg	°E Longitude	°N Latitude	N	k	dp	dm
			Before Structural Correction					
HPT	35.6	22.2	218.3	52.2	62	31.1	1.8	3.5
HT	60.9	6.0	209.0	26.6	26	18.1	3.4	6.9
			After Structural Correction					
HPT	50.4	17.1	209.8	38.5	62	33.5	1.7	3.3
HT	49.4	14.9	211.8	38.7	26	22.4	3.2	6.3
Normal	51.2	19.0	208.2	38.3	37	29.2	2.4	4.6
Reversed	229.2	-14.3	212.1	38.6	25	44.8	2.3	4.5
All determined	50.7	12.2	212.3	36.8	189	7.8	2.0	4.0
Univect. only	50.4	17.1	209.8	38.5	62	33.5	1.7	3.3

See Table 1 footnote.

in the regional structural pattern, expressed as a bend in the Late Cretaceous-early Tertiary fold axes and outcrops (Figure 1). The arm south of this bend strikes N3°E, and the northern arm strike N30°E. The strike of the beds sampled at HPT/HT was consistently N40°E, 10° clockwise to the trend of the northern arm. The deviation of magnetic directions is in the same sense as the structural deflection; a 15° counterclockwise rotation about a horizontal axis would bring the Hechuan poles into coincidence with the other four P/Tr poles (Figure 10). Although the direction of polar discordance coincides with the structural deflection, the magnitude does not (15° versus 30° of structural deflection), possibly suggesting an undetected plunge element. Nonetheless, we are forced to conclude that the Hechuan sites were rotated post-latest Cretaceous/earliest Tertiary folding, in a deformation episode that "bent" the Sichuan basin sedimentary cover around a semi-vertical axis.

The Late Paleozoic-Mesozoic APW Curve

Late Permian-Early Triassic

The Upper Permian Changsing Limestone pole of Lin and others [Lin, 1984; Lin et al., 1985] (3 in Figure 10) is approximately equivalent in age to the GPW and LP poles of this study and to pole 1. However, pole 3 is considerably displaced from these statistically indistinguishable poles and from the northward trend exhibited by South China Block paleopoles (Figure 10). Both we and Heller et al. [1988] experienced extreme difficulty in obtaining univectorial directions from Upper Permian limestones in South China. Lin [1984] apparently also experienced difficulty: except for a single sample [Lin, 1984, Figure A18], none of the data that he obtained are displayed. Much like our samples, the demagnetization behavior of the one example he showed is erratic and not linear, much less univectorial to the origin. Pole 3 was derived from only 13 stratigraphic horizons, and all were of only normal polarity [Lin, 1984]. From the same time interval, we obtained with about equal frequency, both normal and reversed polarity. These considerations and the inconsistent position relative to the other poles suggest that pole 3 may not be a representative pole for this time.

The restudy of the Late Permian-Early Triassic Emeishan Basalt [Zhao and Coe, 1987] led to a recalculated pole which combined their new data with a careful selection of the previously published data. This pole, 4 in Figure 10, also is significantly displaced from the Late Permian-Early Triassic group of poles. The Emeishan Basalt is dated at about 236 m.y. and has mixed polarity [Zhao and Coe, 1987]. This age corresponds to the Early Triassic [Armstrong, 1978; Webb, 1981, Odin and Letolle, 1982; Harland et al., 1982; Haq et al., 1987] and the mixed polarity also indicates a latest Permian to Early Triassic age (discussed later). The pole therefore should be time equivalent to the GPW, LP, and 1 poles; the reason for its significant discordance is unclear, although significant rotation of the Emei Mountain area is a possibility [Z. Zhang, unpublished data,1984].

Despite the discrepancies posed by the Emeishan Basalt and Changsing Limestone poles, the statistically identical nature of poles, 1, GPW and LP, as well as 2, suggests that their position is the location of the latest Permian and Early Triassic pole for south China, and perhaps for all of the Triassic as well. We therefore have calculated a mean latest Permian/Triassic pole for this block, listed in Table 3. Assuming a 15° rotation, a second mean pole including the HPT/HT data also was calculated but, although identical, is less reliable because the amount of rotation of the Hechuan sites is uncertain.

Triassic APW

Several investigators have have suggested scenarios in which the North and South China blocks were positioned close to each other in the Permian, involved in collision in the Early

TABLE 3. Mean Pole for Latest Permian and Triassic of South China

Poles	°E Longitude	°N Latitude	N	R	k	a95
GPW,LP,1,2	223.6	46.1	4	3.997	1161.2	2.7
(GPW,LP,1,2,HPT/HT	223.1	46.6	5	4.996	1103.5	2.3)

Triassic, and sutured during the Triassic and part of the Jurassic [Lin et al., 1985; Zhao and Coe, 1987]. APW so generated may be visible in the pole positions of this study, particularly in the sequence of directions within the lengthy HPT section. HPT spans 1000 m without (as will be discussed in the magnetostratigraphy section) hiatuses of significance, thus representing most of Early Triassic time.

Zhao and Coe's [1987] suturing model involved a scissorlike closure between the North and South China blocks. They postulated a point collision in the Early Triassic and closure during the Triassic and part of the Jurassic. The Hechuan section (HPT) is particularly well situated to address this Triassic suturing hypothesis, as it is distantly located with respect to their inferred hinge point; thus the motion that they propose would translate to largely northward motion of this site. APW generated by the scissor closure therefore would be particularly visible at this section. Hence the data of HPT were examined for APW, using two different assessment techniques. First, virtual poles were calculated for each sample. As illustrated by Figure 11, no significant trend, can be observed throughout the stratigraphic section, either in category 1 samples (univectorial to the origin) (Figure 11a), nor in all samples for which it was possible to interpret polarity (categories 1-3) (Figure 11b).

The second method involved calculating separate poles from the two Early Triassic formations overlying one another. These pole positions are marginally statistically distinct (Table 4), and the very small difference is in the correct direction (assuming rotation of the Hechuan site) to be a reflection of APW. However, further examination of the sample population shows that this difference is an artifact of a few samples, and possibly a secondary magnetization phenomenon. The samples contributing to the two poles were separated into respective normal and reversed groups, and four new poles were calculated. It is obvious that only the pole of the normal samples of the Jialingjiang Formation is significantly displaced; the other three poles are statistically identical. The following reasoning dictates dismissal of the concept of APW between the time of the two formations: the reversed sample directions from the Jialingjiang Formation agree with the normal and reversed directions of the underlying Feixianguan strata; the observed overprints in the HPT section are exclusively of normal polarity; the discordant normal samples come from a restricted portion of the dolomitic upper part of the Jialingjiang Formation; and this upper part of the

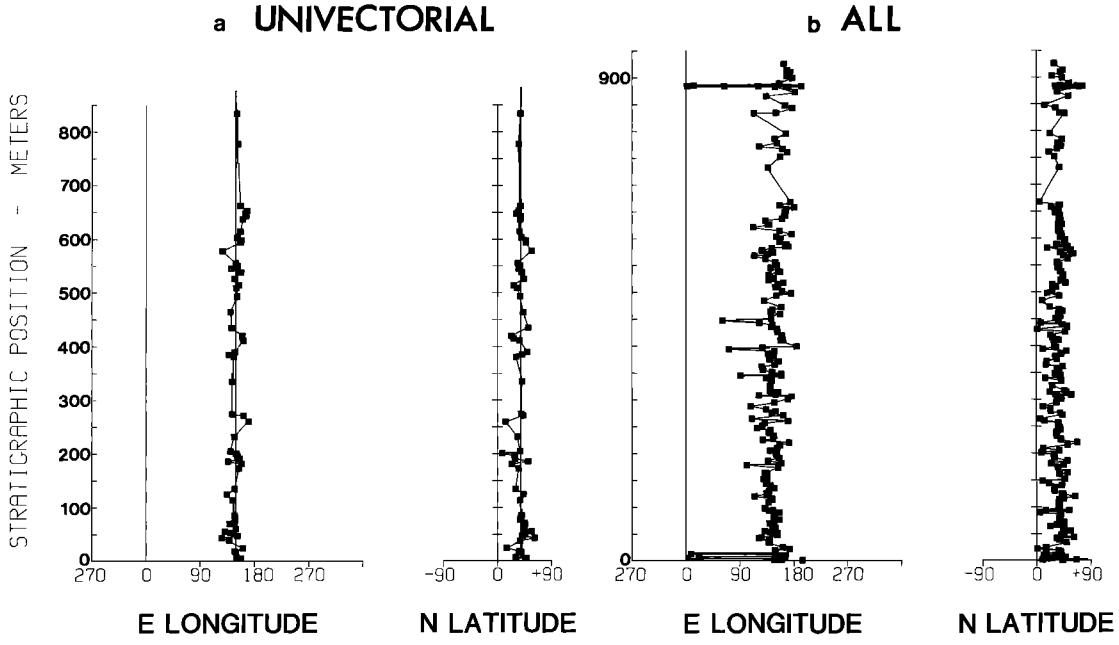


Fig. 11. Stratigraphic plot of pole positions calculated from individual samples at the Hechuan locality. (a) Only samples which were univectorial to the origin; (b) all samples for which polarity could be determined or inferred.

TABLE 4. Early Triassic Formation Poles

Formation	Declination,deg	Inclination,deg	°E Longitude	°N Latitude	N	k	dp	dm
Feixianguan	49.4	14.0	212.2	38.5	36	30.0	2.3	4.5
Jialingjiang	51.8	21.2	206.4	38.4	26	47.5	2.3	4.4
Feixianguan R	230.0	-12.9	212.4	37.6	13	33.4	3.8	7.4
Feixianguan N	49.0	14.6	212.1	39.0	23	27.4	3.1	6.0
Jialingjiang R	228.4	-15.8	211.9	39.8	12	66.0	2.8	5.5
Jialingjiang N	55.0	25.8	201.6	37.0	14	57.4	3.1	5.7

formation exhibits a high incidence of remagnetization.

We conclude that position of the pole did not change through Feixianguan and Jialingjiang time, implying that no measurable movement of the South China Block occured during the Early Triassic. Lack of APW during the entire Triassic is suggested by the coincidence of the Late Triassic paleopole with those of the Early Triassic.

Post-Early Triassic APW

A Middle Triassic paleopole (5 [Lin et al., 1985]) lies along the general northward path of APW for South China. Another Middle Triassic pole (6 [Chan et al., 1984]) lies near the Jura-Cretaceous poles. Pole 6 was calculated from an unacceptably small number of samples (eight); further, its close proximity to the Jura-Cretaceous poles suggests possible remagnetization. Pole 5 was also calculated from a very small sample population (11); its position may suggest APW during the Triassic, but such evidence is at odds with the coincidence of the Late Triassic paleopole, 2, with those of the Early Triassic. It can be argued that pole 2 is also derived from a small sample population (23) or that undetected rotations could conceivably account for its coincident position in the Early Triassic pole group. However, the sample population of pole 2 is marginally larger than that of either of those of the Middle Triassic poles. The absence of APW in the Early Triassic HPT record is congruent with, although does not prove, a lack of APW between Early and Late Triassic implied by the position of pole 2.

Suturing of the China Blocks

With the addition of several new data, the APW path for South China now indicates primarily northward motion of the block from late Paleozoic through Tertiary time (Figure 10). A significant pause in polar motion is indicated during Late Permian through Late Triassic time. Absence of APW during the Triassic suggests either a later assembly of the South and North China blocks or a joining due to the motion of the North China Block (NCB). Southward motion of the NCB from the Triassic to the Jurassic can be inferred from the sparsely defined APW path shown by Lin at al. [1985].

The absence of APW in the entire Early Triassic shown by this study corroborates Opdyke et al.'s [1986] suggestion that plate motion involving the final suturing of the North and South China blocks was post-Triassic. South China, North China, Tarim, the Russian platform, and western Europe have the same pole position by Late Cretaceous time [Achache et al., 1983; Kent et al., 1986; Li et al., 1987]. The statistically identical pole positions for the North and South China blocks from Middle Jurassic through Late Cretaceous time [Courtillot and Besse, 1986] may be a real stillstand of the block or may be a remagnetization phenomena. The post-Triassic suturing inferred from this study then implies that suturing occurred in the Jurassic or in the Cretaceous.

Magnetic Polarity in the Late Permian and Early Triassic

Correlation of the Sichuan Sections

Figure 12 compares the polarity interpretations of the four sections. The same polarity pattern is exhibited by both the HPT and HT sections, implying the validity of this part of the polarity record and suggesting that true earliest Triassic geomagnetic field behavior was recorded.

All sections indicate that the lowest Triassic strata were of normal polarity. Earliest Triassic magnetic polarity appears to consist of simple pattern of normal followed by reversed polarity. The northern and southern Sichuan sections indicate the same magnetic polarity pattern, but the significant differences in thicknesses of the magnetic and biostratigraphic patterns imply that the Feixianguan Formation is approximately twice as thick at HPT as it is at GPW. The two localities are separated by about 700 km, and the increasingly terrigeneous content of the Feixianguan Formation to the south makes such a lateral variation in sedimentation rate plausible. Hence the lengths of the two columns in Figure 12 have been scaled so as to match the lowest Triassic polarity boundary.

The magnetic polarity of our two sections of Late Permian age, LP and GPW, appear to be contradictory (Figure 12). While section LP exhibits normal polarity in the earliest Triassic in agreement with the other sections, it displays a relatively lengthy normal polarity interval in the youngest part of the Late Permian, in contrast to the indications of reversed polarity at GPW (Figure 12). However, the Permo-Triassic boundary at LP is a fault contact; hence it is likely that there has been tectonic removal of the top of the Permian section at LP. Other explanations are also possible: differences in sedimentation rate,

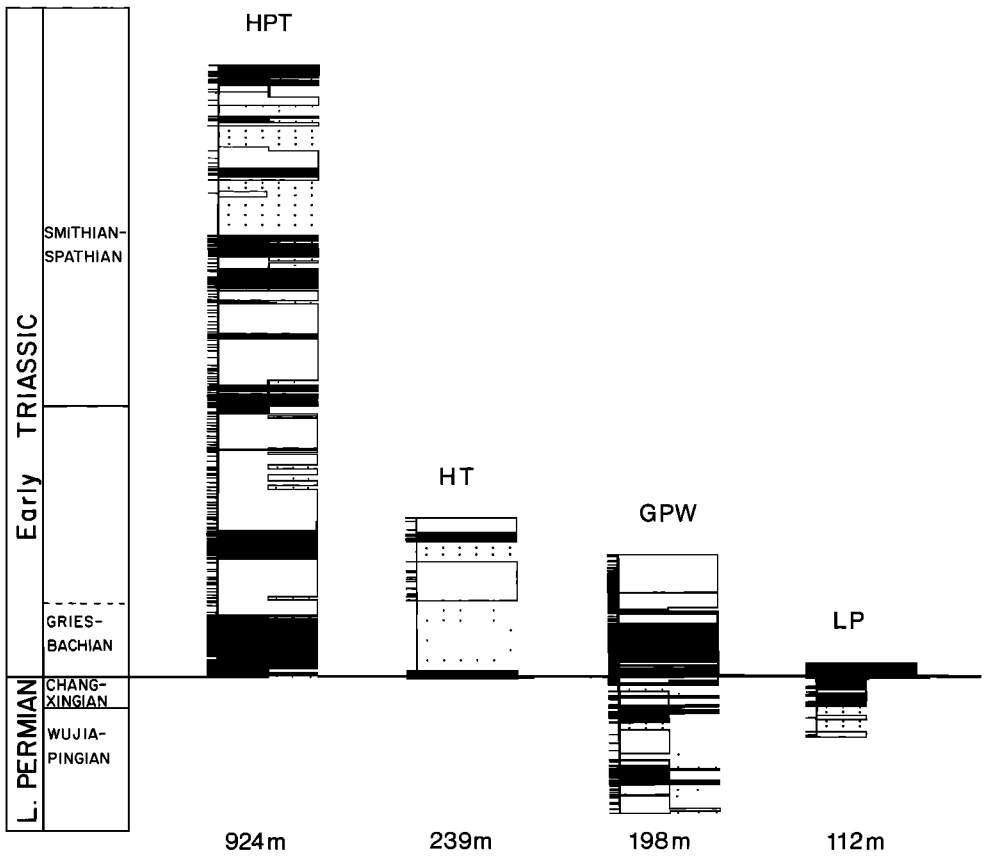


Fig. 12. Comparison of the magnetic polarity interpretations from each locality. HPT, HT, and LP are plotted at the same scale, while GPW is shown at twice that scale (see text).

hiatuses, or inaccuracy of the inferred polarity record from GPW. But we show below that the record from the GPW section is similar to other Late Permian polarity records. The tectonic nature of the P/Tr boundary at the LP section inclines us to ignore its polarity implications.

Correlation With Published Magnetic Polarity Sequences

Permian. The Permian encompassed a lengthy period of almost exclusively reversed polarity, terminated finally by the mixed polarity Illawarra Interval [McElhinny and Burek, 1971], beginning in the Late Permian and continuing through the Early Triassic. Worldwide studies have shown the presence of intervals of normal polarity in the Late Permian. Menning [1986] conducted a global assessment of Late Permian magnetic polarity. His correlations, however, were based on a little used biostratigraphic correlation scheme [Anderson, 1980], the basis of which is unpublished. Nevertheless, he concluded that the upper 21% of the Permian was a time of mixed polarity. Only the percentage of Permian time having mixed polarity was determined; the pattern of magnetic polarity was not evaluated.

Late Permian magnetic polarity correlations are necessarily imprecise, as no geologic standard section exists for this time. Great diversity of fauna apparently existed around the globe, and every region had a different fauna and thus has different stage names. Accordingly, comparing different magnetostratigraphic sections and estimating their time duration is quite difficult. Moreover, many Late Permian/Early Triassic magnetic polarity sequences were determined from terrestrial rocks possessing meager faunal content. Figure 13, a composite of published magnetic polarity records and our results, represents the current state of our knowledge of magnetic polarity in the Late Permian and Early Triassic.

Figure 13 includes all studies which reported magnetostratigraphic results tied to some amount of biostratigraphy. All polarity interpretations and ages presented in Figure 13 are those published by the original authors. The heights of the columns reflect the appropriate length of Late Permian or Early Triassic time spanned by the magnetostratigraphic sections, based on the ages indicated by the authors of the studies. The ages associated with the Soviet magnetostratigraphic sections are particularly uncertain as investigators in the USSR use magnetic polarity to determine age in the entirely terrestrial strata which their studies encompass. In particular, they use of the presence of normal polarity intervals in Late Permian to define the late Tartarian and thus to date the strata. Hence the reliability of the Soviet records as independent indicators of geomagnetic polarity in any part of this time interval is unclear.

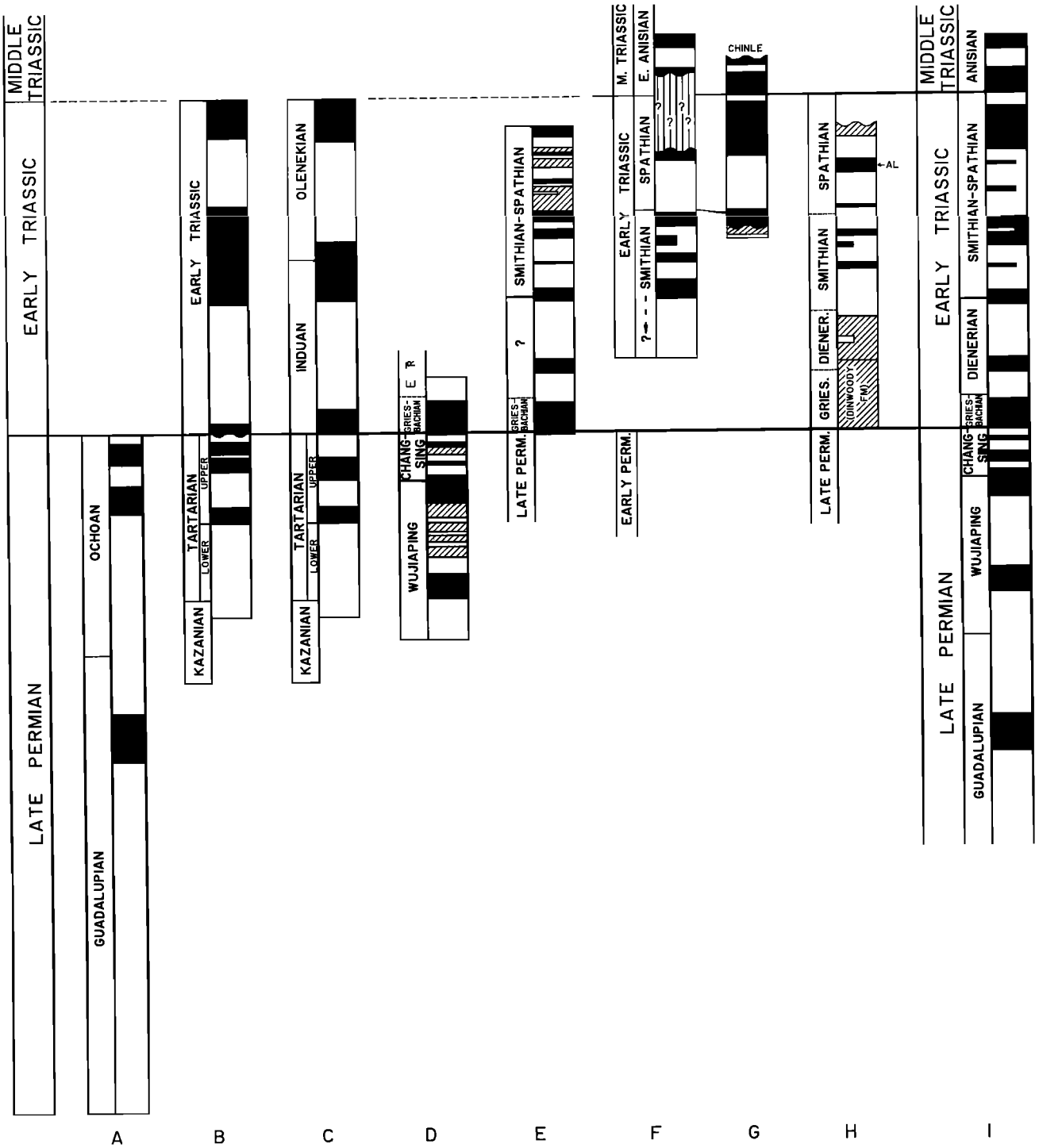


Fig. 13. Global comparison of polarity studies across the Permian-Triassic boundary. A, Late Permian of North America [Nairn and Peterson, 1973]; B, composite of Molostovsky's [1983] Permo-Triassic data from the eastern USSR; C, data of Gurevich and Slautsitays [1985] from the USSR; D, Permian data of this study (GPW locality); E, the Triassic data of this study (HPT locality); F and G, Moenkopi Formation of North America; F represents the eastern part of the depositional basin [Helsley and Steiner, 1974; Lienert and Helsley, 1981; C.E. Helsley, unpublished data, 1972] while G shows the western portion [Purucker et al., 1980; E.M. Shoemaker, unpublished data, 1972]; H, composite of North American Chugwater Group [Picard, 1964; Shive et al., 1984; Herrero-Brevera and Helsley, 1984]; I, composite Late Permian-Early Triassic magnetic polarity sequence.

A study of back reef marine and terrestrial rocks from North America [Nairn and Peterson, 1973] also indicates two normal polarity intervals in the uppermost Permian (A, Figure 13) and suggests a third interval considerably older, in the Capitanian Stage of the Guadalupian Series of the Late Permian. Unfortunately, these data were not obtained from a continuous magnetostratigraphic section but from a few sites separated stratigraphically and geographically. However, the paleomagnetic data indicative of the normal polarity intervals appear sound [Peterson and Nairn, 1971], particularly those of the Capitanian and latest Ochoan, and the sites have some biostratigraphic control.

Even though the ages of the USSR sections are unclear, numerous studies have been reported (e.g., Molostovskiy [1983] and Gurevich and Slautsitays [1985] columns B/C, Figure 13) and usually indicate the presence of at least two (sometimes three) intervals of normal polarity in latest Permian time.

The magnetostratigraphic record of the Late Permian obtained in this study (GPW section, column D) is similar to the published data, in that we infer at least two, possibly four, normal polarity intervals in the latest Permian. Heller et al. [1988] also obtained generally the same pattern of polarity for this same interval; because of our higher sampling density and detail of results, we have shown our data in Figure 13. Section LP of this study supports the existance of at least one normal polarity interval in the Late Permian. Normal polarity was also reported from northwestern China near the top of the Tartarian [McFadden et al., 1988], but the biostratigraphy is poorly defined. The collective data indicate that the geomagnetic field had at least two, and perhaps more, intervals of normal polarity in Late Permian time.

The GPW section of our study is considered to be one of the most complete records of Late Permian/Early Triassic time. The Permian portion ostensibly represents the entire Late Permian time. The relative thiness (104 m) seems contradictory, and the conflict is likely one of reconciling Chinese and global Permian terminology. Although the Wujiaping and Changsing stages constitute the entire Late Permian in China [Li et al., 1986; Wang et al., 1987], they appear to correspond to only the upper 1/2 to 2/3 of the Late Permian elsewhere [e.g., Zhang, 1984; Wang et al., 1987; Clark and Wang, 1988]. The distribution of normal polarity magnetization within the GPW section is similar to that of the late Tartarian and Ochoan of other sections, corroborating the concept that the Late Permian of the GPW section is equivalent only to the upper part of the Late Permian elsewhere.

Several published studies of Late Permian/Early Triassic rocks are not included in Figure 13. Terrestrial redbed sections of this approximate age in South America have been studied [Valencio, 1979], but age control is insufficient even to permit the identification of the Permian-Triassic boundary, much less a comparison of magnetic polarity with the rest of the world. Stratigraphic sections of Late Permian rocks also have been sampled in Kashmir and Australia [Facer, 1981; Klootwijk et al., 1983]; however, the relationship of magnetic polarity to stratigraphic position or biostratigraphic age was not presented.

Triassic. The Early Triassic has long been indicated to have been a time of mixed polarity, although all paleomagnetic studies have been in terrestrial redbed strata [Khramov, 1963; Picard, 1964; Burek, 1967, Helsley, 1969; Molostovskiy, 1970, 1983; Helsley and Steiner, 1974; Khramov et al., 1974; Slautsitays, 1975; Lienert and Helsley, 1980; Purucker et al., 1980; Shive et al., 1984; Gurevich and Slautsitays, 1985]. All investigations of North American stratigraphic sections indicate that frequent polarity changes, as many as a dozen, occurred during this time (columns F, G, and H of Figure 13). Studies from the USSR, on the other hand, consistently show only five, relatively longer, polarity intervals (e.g. columns B, C). Sampling interval might be the explanation of this discrepancy.

The North American records are the results of studies of the Moenkopi and Chugwater formations. The Moenkopi Formation is shown in columns G and H, which represent composites of more than 17 replicate sections spread laterally across 450 square kilometers [E. Shoemaker, C. Helsley, published and unpublished data]. The two columns represent the eastern, near-source (G) and western, sea margin - tidal flat (H) parts of the basin. Smithian and Spathian ammonites occur sparsely within the formation, especially in the western part of the depositional basin where two marine transgressions interfinger with the redbeds. The Smithian/Spathian boundary is constrained by the presence of the Meekoceras and Tirolites genera, and its position within the magnetic polarity records [E. Shoemaker, pers. comm., 1987] is shown on the columns.

The polarity record from the Chugwater Group (column H) has poorer biostratigraphy than that of the Moenkopi Formation. At its base it intertongues with the marine Dinwoody Formation, which contains late Griesbachian conodonts in the lower part. The upper part of the Dinwoody is devoid of fossils, thus the base of the Chugwater Group is not well constrained, but is probably within the early Dienerian. In the upper part of the Chugwater Group a laterally extensive, few meters thick limestone (Alcova Limestone) occurs. The Alcova Limestone ("AL" on column H) has no age diagnostic faunal elements itself, but has been traced laterally into the upper part of the marine Thaynes Formation [High and Picard, 1969] which has age diagnostic conodonts [Carr and Paull, 1983]. The predominance of reversed polarity with intervening small normal intervals and its inferred age correspond well to the predominantly reversed portion of the Moenkopi of approximately the same age (column G) and to the China data of this study (column E).

The data obtained in this study, sections HPT and HT (column E), bear remarkable resemblance to the frequently alternating polarity of the North American studies. There are two striking consequences of this similarity of polarity records. First, The agreement of these three data sets implies that we have obtained a true record of geomagnetic field history for the Early Triassic. Second, the polarity agreement is remarkable as HPT/HT are marine strata while the North American studies are terrestrial redbeds. One would expect marine strata to have fewer hiatuses of section and to represent a more complete time record. The similarity of the redbed polarity record to that of the marine record suggests that the redbeds recorded geomagnetic field polarity at

the time of their deposition, and that they do not contain substantial hiatuses in deposition.

The new Early Triassic magnetic polarity sequence from China, and its agreement with existing records of the same time, indicate that geomagnetic polarity history in the Early Triassic was a series of eleven major polarity intervals, and with a slight bias toward reversed polarity. As many as 17 intervals are suggested by the records of this study, although this has yet to be verified. The length of the Early Triassic is not well known, indicated to be 5 m.y. [Armstrong, 1978, 1982; Webb, 1981; Odin et al., 1982; Harland et al., 1982] or 10 m.y. [Haq et al., 1987]. The uncertainty in the duration of the Early Triassic makes an estimate of Early Triassic reversal frequency tentative, but suggests between 1 and 3 reversals per million years. What is certain is that the Early Triassic was a time of relatively frequent reversals of the geomagnetic field. The HPT record from this study is currently the best available combination of magnetostratigraphy and definitive biostratigraphy. The agreement of North American redbed sequences and the Chinese section strongly suggests that we have a valid representation of Early Triassic geomagnetic polarity.

Conclusions

More than 1100 m of Late Permian and Early Triassic stratigraphic section have yielded a magnetic polarity sequence within fossiliferous marine strata. The stability of magnetization in the Triassic samples resulted in a well-defined Early Triassic record, while large amounts of secondary magnetization rendered the Late Permian record much more tentative. Pole positions, largely from the Early Triassic samples, corroborate the suggestion that suturing of the North and South China blocks was post-Triassic and therefore must have occurred in the Jurassic or Cretaceous. Both the Late Permian and the Early Triassic magnetic polarity records agree well with published data and allow the construction of a magnetic polarity time scale for this time. At least two, but probably three or more, normal polarity intervals occurred in the Late Permian (Figure 13). The Early Triassic was a time of frequent polarity changes; at least 11 and possibly as many as 17, normal and reversed intervals are indicated. The Permo-Triassic boundary occurs within a normal interval, but since the boundary is always a lithologic change, time is likely to be missing, and thus one or more polarity intervals could be missing. The agreement of the Chinese marine and American redbed sections implies an early acquisition of magnetization in some redbed sequences and moreover a validity of their recorded polarity histories. The close agreement of the marine and terrestrial polarity patterns implies that we have derived a relatively accurate representation of geomagnetic field history. The polarity sequence and associated biostratigraphy of this study form the basis of an Early Triassic magnetic polarity time scale.

Acknowledgments. We thank Don Boyd for suggesting a magnetostratigraphic study of the Chinese Permo-Triassic boundary sections. We are grateful to Yang Zunyi for arranging this joint U.S.-China cooperative project. Our work was supported by NSF grant EAR 84-19125 and the Institute of Geology, Chinese Academy of Geology. We thank M. Fuller for kindly providing a copy of J. Lin's dissertation. Suggestions from several anonomous reviewers greatly improved the manuscript.

References

Achache, J., V. Courtillot, and J. Besse, Paleomagnetic constraints on the Late Cretaceous and Cenozoic tectonics of southeastern Asia, Earth Planet. Sci. Lett., 63, 123-136, 1983.

Anderson, J.M., World Permo-Triassic correlations: Their biostratigraphic basis, in Godwana Five, edited by M.M. Cresswell and P. Vella, pp. 3-17, Balkema, Rotterdam, 1980.

Armstrong, R.L., The Triassic time scale, Spec. Publ. Am. Assoc. Petro. Geol., 6, 73-91, 1978.

Armstrong, R.L., Late Triassic-Early Jurassic time-scale calibration in British Columbia, Canada, in Numerical Dating in Stratigraphy, edited by G.S. Odin, pp. 509-514, John Wiley, N.Y., 1982.

Burek, P.J., Korrelation revers magnetisierter Gesteinfolgen im Oberen Buntsandstein SW-Deutschlands (Correlation of reversely magnetized rock sequences in the Upper Buntsandstein, SW-Germany), Geol. Jahrb., 84, 591-616, 1967.

Carr, T.R., and R.K. Paull, Early Triassic stratigraphy and paleography of the Cordilleran miogeocline, in Meso. Paleogeogr. of West Central United States, edited by M.W. Reynolds and E.D. Dolly, pp. 39-55, Rocky Mountain Sect., Soc. Econ. Paleontol. and Mineral., Denver, 1983.

Chan, L.S., C.Y. Wang, and X.Y. Wu, Paleomagnetic results from some Permian-Triassic rocks from southwestern China, Geophys. Res. Lett., 11, 1157-1160, 1984.

Clark, D.L., and C.-Y. Wang, Permian Neogondolellids from south China: Significance for evolution of the Serrata and Carinata groups in North America, J. Paleontol. 62, 132-138, 1988.

Clark, D.L., C.-Y. Wang, C.J. Orth, and J.S. Gilmore, Conodont survival and low iridium abundances across the Permian-Triassic boundary in south China, Science, 233, 984-986, 1986.

Courtillot, V., and J. Besse, Mesozoic and Cenozoic evolution of the North and South China blocks, Nature, 320, 86-87, 1986.

Facer, R.A., Paleomagnetic data from Permian and Triassic rocks from drill holes in the southern Sydney Basin, New South Wales, Tectonophysics, 74, 305-321, 1981.

Gurevich, Ye.L., and I.P. Slautsitays, A paleomagnetic section in the Upper Permian and Triassic deposits on Novaya Zemlya, Int. Geol. Rev., 27, 168-177, 1985.

Haq, B., J. Hardenbol, and P. Vail, Chronology of fluctuating sea levels since the Triassic, Science, 235, 1156-1167, 1987.

Harland, W.B., A.V. Cox, P.G.Llewelyn, C.A.G. Pickton, C.A.G. Smith, and R. Walter, A Geologic Time Scale, 131 pp., Cambridge University Press, Cambridge, 1982.

Heller, F., W. Lowrie, Li H., and Wang J., Magnetostratigraphy of the Permo-Trissic boundary section at Shangsi, Earth Planet. Sci. Lett., 88, 1988, 348-356.

Helsley, C.E., Magnetic reversal stratigraphy of the Lower Triassic Moenkopi Formation of western Colorado, Geol. Soc. Am. Bull., 80, 2431-2450, 1969.

Helsley, C.E., and M.B. Steiner, Paleomagnetism of the Lower Triassic Moenkopi Formation of western Colorado, Geol. Soc. Am. Bull., 85, 457-464, 1974.

Herrero-Brevera, E., and C.E. Helsley, Paleomagnetism of a polarity transition in the Lower (?) Triassic Chugwater Formation, Wyoming, J. Geophys. Res., 88, 3506-3522, 1983.

High, L.R., and M.D. Picard, Stratigraphic relations within the upper Chugwater Group (Triassic), Wyoming, Amer. Assoc. Pet. Geol. Bull., 53, 1091-1104, 1969.

Kent, D.V., G. Xu, K. Huang, W.Y. Zhang, and N.D. Opdyke, Paleomagnetism of Upper Cretaceous rocks from south China, Earth Planet. Sci. Lett., 79, 179-184, 1986.

Khramov, A.N., Paleomagnitoe izuchenie razrezov verkhnie permi i nizhnego triasa severa i vostoka Russkoi platformy, in Paleomagnitnye stratigraficheskie issledovania, pp. 145-174, Gostoptekhizdat, Leningrad, 1963.

Khramov, A.N. (Ed.), Paleomagnetism paleozoya, 236 pp., Nedra, Leningrad, 1974.

Kirschvink, J.L., The least squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc., 62, 699-718, 1980.

Klootwijk, C.T., S.K. Shah, J. Gergan, M.L. Sharma, B. Tirkey, and B.K. Gupta, A paleomagnetic reconnaissance of Kashmir, northwestern Himilaya, India, Earth Planet. Sci. Lett., 63, 305-324, 1983.

Kozur, H., Beitrage zur Stratigraphie des Perms, teil II, Die Conodonten chronologie des Perms, Freiberg. Forschungsh. R C, 334, 85-161, 1978.

Kummel, B., and C. Teichert, The Permian-Triassic boundary in central Tethys, Mem. Can. Soc. Pet. Geol., 2, 17-34, 1973.

Li, Y.-P., Zhang, Z., M. McWilliams, A. Cox, R. Sharps, Y. Zhai, Y. Li, and Q. Li, Mesozoic results from the Tarim craton, China, Eos Trans. AGU, 68, 1259, 1987.

Li, Z., et al., Mass extinction and geologic events between Palaeozoic and Mesozoic Era, 26 pp., 5 plates, Minist. of Geol., Chin. Acad. of Geol. Sci., Beijing, 1986.

Lienert, B.R., and C.E. Helsley, Magnetostratigraphy of the Moenkopi Formation at Bears Ears, Utah, J. Geophys. Res., 85, 1475-1480, 1980.

Lin, J.-L., The apparent polar wander paths for the North and the South China blocks, Ph.D. dissertation., Univ. of Calif., Santa Barbara, 1984.

Lin, J.-L., M. Fuller, and W.-Y. Zhang, Preliminary Phanerozoic polar wander paths for the North and South China blocks, Nature, 313, 444-449, 1985.

McElhinny, M.W., and P.J. Burek, Mesozoic paleomagnetic stratigraphy, Nature, 232, 98-102, 1971.

Menning, M., Zur Dauer des Zechsteins aus magnetostratigraphischer Sicht, Z. Geol. Wiss., 14, 395-404, 1986.

McFadden, P.L., X.H. Ma, M.W. McElhinny, and Z.K. Zhang, Permo-Triassic magnetostratigraphy in China: northern Tarim, Earth Planet. Sci. Lett., 87, 152-160, 1988.

Molostovskiy, E.A., Paleomagnetic stratigraphy of the Upper Permian and Triassic of the eastern European part of the USSR (in Russian), 167 pp., Univ. of Saratov, Saratov, USSR, 1983.

Nairn, A.E., and D.N. Peterson, A review of Permian and Triassic paleomagnetic data with respect to paleogeographic conditions at, and the location of, the Permian-Triassic boundary, Mem. Can. Soc. Pet. Geol., 2, 694-713, 1973.

Nakazawa, K., Y. Bando, and T. Matsuda, The Otoceras woodwardi zone and the time-gap at the Permian-Triassic boundary in east Asia, in Symposium Tsukuba '78, Geology and Paleontology of Southeast Asia, edited by T. Kobayashi et al., pp. 75-90, University of Tokyo Press, Tokyo, 1980.

Newell, N.D., The search for a Paleozoic-Mesozoic boundary stratotype, Schriftenr. Edwiss. Komm. Osterr. Akad. Wiss., 4, 9-19, 1978.

Odin, G.S., and R. Letolle, The Triassic timescale in 1981, in Numerical Dating in Stratigraphy, edited by G.S. Odin, pp. 523-533, John Wiley, New York, 1982.

Olsen, E.C., Extinctions of Permian and Triassic nonmarine vertebrates, Spec. Pap. Geol. Soc. Am., 190, 501-516, 1982.

Opdyke, N.D., K. Huang, G. Xu, and W.Y. Zhang, Paleomagnetic results from the Triassic of the Yangtze Platform, J. Geophys. Res, 91, 9553-9568, 1986.

Otofuji, Y., Y.O. Jin, T. Hirajima, D.M. Kyung, and S. Sasajima, in The Tectonic and Geologic Evolution of Southeast Asian Seas and Islands: part 2, Geophys. Monogr. Ser., vol. 27, edited by D.E. Hayes, pp. 388-396, AGU, Washington, D.C., 1983.

Peterson, D.N., and A.E.M. Nairn, Paleomagnetism of Permian redbeds from the southwestern United States, Geophys. J. R. Astron. Soc., 23, 191-205, 1971.

Picard, M.D., Paleomagnetic correlation of units within the Chugwater (Triassic) Formation, Am. Assoc. Pet. Geol. Bull., 48, 269-291, 1964.

Purucker, M.E., D.P. Elston, and E.M. Shoemaker, Early acquisition of characteristic magnetization in red beds of the Moenkopi Formation (Triassic), Gray Mountains, Arizona, J. Geophys. Res. 85, 997-1012, 1980.

Sepkoski, J.J., Jr., Mass extinctions in the Phanerozoic oceans: A review, Spec. Pap. Geol. Soc. Am., 190, 283-289, 1982.

Shive, P.N., M.B. Steiner, and D.T. Huycke, Magnetostratigraphy, paleomagnetism, and remanence acquisition in the Triassic Chugwater Group of Wyoming, J. Geophys. Res., 89, 1801-1815, 1984.

Slautsitays, I.P., Stratigrapfia permskikh i triasovykh otlozhenyi Mangyshlaka po paleomagnitnym dannym, in Kontinentalnye krasnotzvetnye otlozhenia permi i triasa, pp. 113-115, University of Saratov, Saratov, USSR, 1975.

Sun, Y.-Y., et al., The discovery of iridium in the Permian-Triassic boundary clay in Changxing, Zhejiang, China and its significance, report, 10 pp., IGCP Working Group 183, Beijing, 1984a.

Sun, Y.-Y.,et al., The discovery of iridium in the Permian-Triassic boundary clay in Changxing, Zhejiang, China and its significance, Int. Geol. Congr. Abstr., 27th, 8, 309, 1984b.

Sweet, W.C., Graphic correlation of Permo-Triassic rocks in Kashmir, Pakistan and Iran, Geol. Palaeontol., 13, 239-248, 1979.

Tozer, E.T., The earliest marine Triassic rocks: Their definition, ammonoid fauna, distribution and relationship to underlying formations, Mem. Can. Soc. Pet. Geol. 2, 549-556, 1973.

Tozer, E.T., The significance of the ammonoids Paratirolites and Otoceras in correlating the

Permian-Triassic boundary beds of Iran and the People's Republic of China, Can. J. Earth Sci., 16, 1524-1532, 1979.

Valencio, D.A., Reversals and excursions of the geomagnetic field as defined by palaeomagnetic data from Upper Paleozoic-Lower Mesozoic sediments and igneous rocks from Argentina, Global Reconstruction of the Geomagnetic Field During the Paleozoic, edited by M.W. McElhinny et al., Adv. Earth Planet. Sci., 10, 137-142, 1979.

Van der Voo, R., and J.G. Geissman, Late Permian and Triassic paleopoles for North America: Deviations from the small-circle (PEP) track, EOS Trans. AGU, 69, 338, 1989.

Wang, C.-Y., S.M. Ritter, and D.L. Clark, The Sweetognathus complex in the Permian of China: Implications for evolution and homeomorphy, J. Paleontol., 61, 1047-1057, 1987.

Wang, Z., Z. Li, Z. Xia, X. Liu, and R. Li, Guide to a geological excursion to see the Triassic and Jurassic between Beibei of Chongqing and Hechuan, Sichuan, 47 pp., 23 plates, Chengdu Inst. of Geol. and Miner. Resour., Minist. of Geol., Geol. Bur. of Sichuan Prov., Chengdu, 1979.

Webb, J.A., A radiometric time scale of the Triassic, J. Geol. Soc. Aust., 28, 107-121, 1981.

Xu, D.-Y., S.-L. Ma, Z.-F. Chai, X.-Y. Mao, Y.-Y. Sun, Q.-W. Zhang, and Z.-Z. Yang, Abundance variation of iridium and trace elements at the Permian-Triassic boundary at Shangsi, Guangyuan, China, report, 10 pp., IGCP Working Group 183, Beijing, 1984.

Xu, D.-Y., S.-L. Ma, Z.-F. Chai, X.-Y. Mao, Y.-Y. Sun, Q.-W. Zhang, and Z.-Z. Yang, Abundance variation of iridium and trace elements at the Permian-Triassic boundary at Shangsi in China, Nature, 314, 154-156, 1985.

Yang, Z., Y. Cheng, and H. Wang, The Geology of China, Clarendom Press, Oxford, 1986.

Zhang, Z.-Q., The Permian System in south China, Newsl. Stratigr., 13, 156-174, 1984.

Zhao, X., and R.S. Coe, Paleomagnetic constraints on the collision and rotation of North and South China, Nature, 327, 141-144, 1987.

Zhu, Q., et al., Field guide to the Permo-Triassic section of Liangfenya, Chongqing, Sichuan Province, 14 pp., Geol. Dep. Chongqing Petrol. Sch., Chongqing, 1984.

J. Ogg, Department of Earth and Atmospheric Sciences, Purdue University, West Lafayette, IN 47907.

M. Steiner, Department of Geology and Geophysics, University of Wyoming, Laramie, WY 82071.

S. Sun and Z. Zhang, Institute of Geological Sciences, Chinese Academy of Geological Sciences, Beijing, People's Republic of China.

(Received May 25, 1988;
Revised November 30, 1988;
accepted January 4, 1989.)